

# Akshith Macharla

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## Education

### California State University, Chico

Master of Science in Computer Science – GPA 3.95/4.00

Chico, CA

Expected Dec 2026

Coursework: Operating Systems, Data Structures and Algorithms, Software Design, Web Development, Machine Learning

### Chaitanya Bharathi Institute of Technology

Bachelor of Engineering in Electrical and Electronics Engineering

Hyderabad, India

2017 – 2021

Coursework: Microprocessors and Microcontrollers, Embedded Systems, Digital Electronics and Logic Design, Digital Signal Processing

## Technical Skills

- **Languages:** Java, Python, C++, SQL, TypeScript, Bash
- **Backend:** Spring Boot, Spring Data JPA / Hibernate, REST APIs, gRPC, Protocol Buffers, FastAPI, Django, Flyway, Maven
- **Data & Messaging:** PostgreSQL, MongoDB, Elasticsearch, Redis, Kafka, RabbitMQ, Spark Structured Streaming, pandas, FAISS
- **Cloud & DevOps:** AWS, Kubernetes, EKS, Docker, GitHub Actions, Jenkins, Azure Container Apps, Linux, Git
- **Testing & Observability:** JUnit, Mockito, Testcontainers, pytest, Postman, Prometheus, Grafana, Dynatrace

## Experience

### Cognizant Technology Solutions

Hyderabad, India

Programmer Analyst / Associate Software Engineer

Dec 2021 – Aug 2024

- Built and maintained Java/Spring Boot and PostgreSQL services for policy lookup, claims intake, and audit history (10K+ users).
- Designed REST API contracts with request validation, structured error responses, and centralized exception handling.
- Implemented business rules for policy status, claim eligibility, required documents, permissions, and workflow state transitions.
- Cut p95 API latency from ~250 ms to ~200 ms by removing redundant reads and optimizing Hibernate/SQL queries.
- Triageed 100+ production incidents (API failures, database inconsistencies, regressions) with Dynatrace, supporting 99.9% uptime.
- Wrote JUnit/Mockito tests, validated APIs in Postman, and supported AWS releases through Maven, Jenkins, and JIRA.

### Edureka

Hyderabad, India

Data Science and Machine Learning Intern

Jan 2023 – Jul 2023

- Built a reproducible Python/pandas and PostgreSQL pipeline to clean, validate, and transform 1.3M+ records.
- Wrote SQL (joins, CTEs, window functions) for cohort and KPI tables; automated data-quality checks and built Tableau dashboards.
- Tuned an Elastic Net logistic regression with time-aware 5-fold cross-validation, raising ROC-AUC from 0.71 to 0.84.

### California State University, Chico

Chico, CA

Web Developer – TypeScript, SvelteKit, Cloudflare Workers

Nov 2025 – Feb 2026

- Shipped a university event site end to end, including a Cloudflare Workers + D1 registration backend with idempotent sign-ups.

## Projects

### Flash-Sale Seat Reservation Service | Java 21, Spring Boot, PostgreSQL, Testcontainers, Docker, Azure

GitHub

- REST API for TTL seat holds, bookings, and releases; GitHub Actions CI/CD deploys Docker images to Azure Container Apps.
- Prevented double-booking in the database, not the app: partial unique indexes allow one active hold and one booking per seat, mapped to 409 Conflict; a Testcontainers test races parallel requests at one seat and asserts exactly one wins.

### Distributed Task Platform | Python, FastAPI, RabbitMQ, PostgreSQL, Prometheus, Grafana, Docker

GitHub

- Async job API with Idempotency-Key dedup; task state in PostgreSQL, jobs through a durable RabbitMQ queue to scaled workers.
- At-least-once processing: workers ACK only after the state commit, retry failures, then route to a dead-letter queue; a 2,000-task load test ran at 31 tasks/s on 1 worker and ~42 on 3.

### Real-Time Funnel Analytics & Anomaly Detection | Kafka API (Redpanda), Spark, PostgreSQL, Grafana

GitHub

- Streamed view, cart, and purchase events through a Kafka-compatible topic into Spark Structured Streaming, computing per-minute funnel metrics with event-time windows, a 30 s watermark, and checkpointed offsets.
- A Python job fits EWMA forecasts with  $3\sigma$  bands and flags anomalous minutes via idempotent ON CONFLICT upserts.

### Cited Document Q&A Service (RAG) | Python, FastAPI, FAISS, sentence-transformers

GitHub

- FastAPI service over 24 NIST PDFs that must cite sources or abstain; 75-question eval harness: 69% pass, 100% citation coverage.

### Parallel BPE Tokenizer | C++17, multithreading, CMake

GitHub

- GPT-2 byte-level BPE with a thread pool and per-thread merge caches: 5.0× scaling on 8 threads, 2.11M tokens/s, 91.6% of tiktoken.

## Open-Source Contributions

- **PyTorch ExecuTorch** (merged #22892): graph pass moving attention scaling after QK matmul so int8 scales fold on Arm Cortex-M.
- **vLLM llm-compressor** (merged #3224): unified per-model fused-layer mappings into one shared resolver; validated on H100.
- **NVIDIA CUTLASS** (#3615, in review): C++/CUDA fix for split-K GEMM silently using wrong scale factors; tests that expose it.

## Certifications & Activities

- **AWS Certified Developer – Associate** | **Cal Hacks 12.0:** Chaos Reviewer, an AI code-review agent with ~7.2K agent interactions